



ENVI 5.6 Track — Atmospheric Correction & Spectral Analysis

GeoCascade Chapter 1 — ENVI Workflow Guide *ENVI 5.6 + IDL | Torres del Paine, Patagonia*

When Do You Need Atmospheric Correction?

[!IMPORTANT] **Never apply FLAASH to data that is already atmospherically corrected.** Applying correction twice will corrupt your reflectance values.

Dataset	Level	Status	Action
Sentinel-2 L2A	BOA Surface Reflectance	✔ Corrected by ESA Sen2Cor	Use directly
Landsat 9 L2SP	Surface Reflectance + ST	✔ Corrected by USGS LaSRC	Use directly
Sentinel-2 L1C	TOA Radiance	✘ Raw, uncorrected	Apply FLAASH or Sen2Cor
Landsat 9 L1TP	TOA Radiance	✘ Raw, uncorrected	Apply FLAASH or DOS1
Copernicus DEM	Elevation model	N/A	Use directly
ERA5 / CHIRPS	Climate reanalysis	N/A	Use directly

The Chapter 1 Python pipeline downloads **L2A and L2SP** — already corrected. The ENVI FLAASH workflow teaches the process for when you acquire raw L1C/L1TP data.

FLAASH Parameters for Torres del Paine

FLAASH uses the **MODTRAN5** radiative transfer model. The key parameters are:

Parameter	Value	Reason
Atmospheric model	Sub-Arctic Summer	Best match for 51°S summer conditions
Aerosol model	Maritime	Pacific onshore winds dominate Patagonian aerosol
Initial visibility	40 km	Typical clear-sky visibility in Torres del Paine
Scene centre lat	−51.0°	Southern Hemisphere (negative sign required)
Scene centre lon	−73.0°	West (negative sign required)
Scene elevation	0.15 km	Mean valley floor elevation
Water retrieval	Enabled	1135 nm absorption feature used

Parameter	Value	Reason
Aerosol retrieval	Enabled	2-band Kaufman-Tanre method

[!NOTE] **Why Sub-Arctic Summer?** Torres del Paine is at 51°S — equivalent in character to sub-Arctic latitudes in the Northern Hemisphere. The Sub-Arctic Summer profile in MODTRAN5 has the best-fit temperature lapse rate, humidity profile, and ozone column for mid-latitude Southern Hemisphere summer conditions.

[!NOTE] **Why Maritime aerosol?** Persistent westerly winds off the Pacific bring clean marine aerosol (sea salt, low loading) to western Patagonia. This differs from inland areas where Rural or Tropospheric aerosol models are more appropriate.

FLAASH radiative transfer equation:

$$L_{at-sensor} = \frac{\rho_{surface} \cdot E_{down} \cdot \cos \theta}{\pi} \cdot T(\lambda) + L_{path}(\lambda)$$

Where:

- $\rho_{surface}$ = surface reflectance (what we solve for)
- E_{down} = downwelling irradiance (modeled by MODTRAN5)
- $T(\lambda)$ = atmospheric transmittance
- L_{path} = path radiance (atmospheric scattering)

Running the IDL Script

01_flaash_correction.pro — recommended for most users.

Step-by-step

1. Open **ENVI 5.6**
2. Prepare your input data (if bands are separate):
 - **Raster Management > Layer Stacking**
 - Add all Landsat L1TP bands (B1–B7)
 - Output format: **ENVI** (BSQ interleave)
 - Save as **landsat9_l1tp_stack.dat**
3. Edit the script — open **01_flaash_correction.pro** in a text editor and update:

```
input_file = 'D:\path\to\landsat9_l1tp_stack.dat'
output_dir = 'D:\path\to\output'
scene_date = '2023-01-15'
```

4. Run the script:
 - **Option A:** ENVI Toolbox → ENVI Modeler → **Run IDL Script** → browse to **.pro**

- **Option B:** IDL Console (at bottom of ENVI): `.compile`
'path\to\01_flaash_correction.pro' then `flaash_patagonia`

5. Monitor progress in the ENVI status bar (5–20 minutes depending on scene size)

Expected outputs

```
data/processed/envi_flaash/
├─ flaash_corrected.dat      ENVI format reflectance
├─ flaash_corrected.hdr     ENVI header
├─ flaash_corrected_rfl.tif  GeoTIFF (ArcGIS Pro / QGIS ready)
├─ flaash_cloud_mask.dat    Cloud mask
├─ flaash_water_vapor.dat   Water vapour column map
└─ flaash_report.txt        FLAASH processing report
```

[!TIP] The `.tif` file is directly loadable in **ArcGIS Pro**, **QGIS**, and Python (`rasterio.open()`). The `.dat` file is ENVI's native format — open it with **File > Open**.

Running the Python API Script

`01_flaash_correction.py` — for automation / batch processing.

ENVI 5.6 includes a built-in Python interpreter with the `envi` module pre-installed.

From ENVI Python Console

1. ENVI → **Tools > Python Console**
2. In the console:

```
exec(open(r'D:\00_AI_Aplications\ClimateChange\Chapter_01\envi\01_flaash_corr
ection.py').read())
```

As a Script

1. Edit `INPUT_FILE`, `OUTPUT_DIR`, and `SCENE_DATE` in the script
2. ENVI → **Tools > Python Script** (if available in your ENVI installation)
3. Select and run the script

[!IMPORTANT] The script gracefully handles missing `envi` module: if you run it outside ENVI, it prints clear instructions and exits without crashing.

Spectral Analysis After FLAASH

After running FLAASH, perform spectral profiling in ENVI:

Manual ROI Method (ENVI GUI)

1. **File > Open** → `flaash_corrected_rfl.tif`
2. In the Layer Manager, right-click the raster → **Open in Display**
3. Draw ROIs over representative areas:
 - Grey Glacier (bright white/cyan in NIR)
 - Lenga beech forest (strong NIR reflectance plateau)
 - Grey Lake open water (low reflectance all bands)
 - Rock outcrop (flat, moderate reflectance)
4. **Tools > Region of Interest (ROI) → ROI Tool**
5. **Tools > Spectra → Spectral Profile** → select your ROIs
6. **File > Export > Spectral Profile to CSV** for use in Chapter 2

Expected spectral signatures

Feature	Blue	Green	Red	NIR	SWIR1
Glacier/snow	High	High	High	High	Low
Forest	Low	Moderate	Low	Very High	Moderate
Open water	Moderate	Moderate	Low	Very Low	Very Low
Rock/bare	Moderate	Moderate	Moderate	Moderate	Moderate

[!TIP] The sharp drop in reflectance from NIR to SWIR1 for glaciers is the physical basis of NDSI: $(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1}) > 0.4$ = ice/snow.

Exporting for ArcGIS Pro

After FLAASH correction:

1. **File > Save As > Save As New File**
2. Choose format: **GeoTIFF**
3. Set output path: `data/processed/envi_flaash/flaash_corrected_rfl.tif`
4. Ensure **Output CRS** matches input (usually `WGS 84 / UTM Zone 19S`)

Or use the IDL script which exports automatically via `ExportRasterToFormat`.

ArcGIS Pro will open it directly:

- **Insert > Add Data > Raster Dataset** → select the `.tif`
- Apply **Stretch** symbology (Minimum-Maximum or 2% Clip)

Method Comparison: FLAASH vs DOS1 vs Pre-corrected L2

Method	Accuracy	Speed	Requires ENVI?	Best for
Pre-corrected L2A/L2SP	Highest (physics-based)	Instant (already done)	No	Production analysis

Method	Accuracy	Speed	Requires ENVI?	Best for
FLAASH	High (MODTRAN physics)	5–20 min per scene	Yes (licensed)	L1C/L1TP inputs
DOS1 (script 03)	Moderate (empirical)	Seconds	No	Quick comparison
Sen2Cor	High (physics-based)	10–30 min	No (free)	Sentinel-2 L1C only

[!NOTE] The Python script `03_atmospheric_correction.py` demonstrates **DOS1** on the Landsat L2SP data (treating it as pseudo-L1 for teaching). The RMSE comparison it prints shows how close DOS1 gets to the official LaSRC product.

Files in This Folder

File	Format	Run from
README_ENVI.md	Markdown	Any viewer
01_flaash_correction.pro	IDL script	ENVI 5.6 IDL Console / Toolbox
01_flaash_correction.py	Python	ENVI Python Console

GeoCascade | Chapter 1 | ENVI 5.6 Track | Study area: Torres del Paine, 51°S